

PROBABILITY (GR 11 AND 12)

Probability of an event occurring and sample space:

The probability of Event A occurring is: $P(A) = \frac{n(A)}{n(S)}$

In general, A is the total number of ways a specific event can occur. S is the total number of possible outcomes for the event.

Theoretical and Experimental probability:

In most cases, it takes a large number of trials before experimental and theoretical probabilities approach the same value.

Notation:

Probability of an event lies between 0 and 1. $A = \{1; 2; 3; 4; 5; 6; 7; 8; 9; 10\}$ however means the elements that are part of Event A.

- $P(A) \rightarrow$ The probability of Event A occurring
- $P(A') \rightarrow$ The probability of Event A **NOT** occurring. It is also known as the complement of A
- $P(A \text{ or } B) = P(A \cup B) \rightarrow$ The probability of A or B occurring. $\cup$ is the symbol for **or** and is also known as **union**.
- $P(A \text{ and } B) = P(A \cap B) \rightarrow$ The probability of A and B occurring. $\cap$ is the symbol for **and** and is also known as **intersection**.
- $P(A|B) \rightarrow$ The probability of A occurring **given** that B has occurred.

Inclusive events:

Two events that can occur at the same time are inclusive.

$$P(A \cap B) \neq 0$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Mutually Exclusive events:

Two events that are mutually exclusive cannot occur at the same time.

$$P(A \cap B) = 0$$

$$P(A \text{ or } B) = P(A) + P(B)$$

Exhaustive events:

Two events A and B are exhaustive if together they cover all the elements of the sample space.

$$P(A \text{ or } B) = 1$$

Complementary events:

Mutually exclusive, exhaustive events are **complementary** events.

$$P(\text{not } A) = P(A') = 1 - P(A)$$

The addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If the events are mutually exclusive then: $P(A \cup B) = P(A) + P(B)$, as $P(A \text{ and } B) = 0$.

Dependent and independent events:

Two events are independent if the outcome of one event does not affect the probability of another event occurring. If the outcome of one event changes the probability that another event occurs, the events are said to be dependent.

In most cases, events that take place with replacement are independent events and those without replacement are dependent events.

The product rule:

Test for independent events: $P(A \text{ and } B) = P(A) \times P(B)$

Test for dependent events: $P(A \text{ and } B) \neq P(A) \times P(B)$ or $P(A \text{ and } B) = P(A) \times P(B|A)$

Remember: $P(A \text{ and } B)$ can be written as $P(A \cap B)$

$P(B|A)$ means the probability of B occurring given that A has occurred

Venn diagrams:

Venn diagrams are a graphical way of representing a sample space and its events.

Example:

In a study 160 Gr 11 learners were asked if they take History, Art or Drama. The result of the study are shown below.

- 80 learners said they take History
 - 70 learners said they take Art
 - 80 learners said they took Drama
 - 42 learners said they take History and Drama
 - 36 learners said they take Art and Drama
 - 136 learners said they take at least one of the subjects
 - 14 said they take all 3 subjects.
- a) Record the above information in a Venn diagram.

- b) How many learners take none of the subjects?
- c) How many learners take History and Art but not Drama?
- d) What is the probability that a learner chosen at random takes at least 2 of the subjects?

Tree diagrams:

When there are 2 or more consecutive events taking place, it is often useful to represent the possible solutions on a tree diagram. Tree diagrams are constructed by showing all possible events. They can be used for dependent or independent events. When dealing with tree diagrams always multiply along the branches and add probabilities moving down branches at the end. Write the probability of an event occurring at the top of the branches and the event at the end of the branch.

Example:

5 Blue balls, 3 purple balls and 7 orange balls are placed in a bag. If 2 balls are then selected without replacement:

- a) Draw a tree diagram to represent the above scenario.

- b) Determine the probability of the 2nd ball being purple.

- c) Determine the probability of selecting 2 blue balls.

Contingency tables:

Contingency tables are statistical tables that show the relationship between 2 or more variables. They are often used to determine whether or not events are independent.

Example: A study was conducted between two airports in order to determine whether or not planes departing late was independent of the airports. Based on the results the contingency table below was drawn.

	AIRPORT A	AIRPORT B	TOTALS
# of planes departed late	450		550
# of planes departed on time		300	
Totals	600	400	

- a) Fill in the missing values in the table.
- b) Determine the probability of a plane departing late.
- c) Determine the probability of a plane departing late from Airport A.
- d) Determine the probability of a plane departing late from Airport B.
- e) A manager at Airport A claims that the probability of a plane departing late is independent of the airport that it departs from. Is the manager correct?
- f) Based on your findings, if any, which airport should a traveller choose to depart from?

The fundamental counting principle:

The fundamental counting principle is a principle used to determine the number of different ways to accomplish different tasks. It states that:

If there are m different ways to perform a task and n different ways to perform a different task, the total number of different ways in which both tasks can be performed is $m \times n$.

Example 1: A local ice-cream parlour has 4 different flavours of ice-cream with a choice of 8 different toppings for dessert. If a customer chooses 1 topping and one flavour, how many different choices of dessert are there?

Example 2: At a matric farewell, the set menu below shows the choices that guests have for starters, mains and dessert. How many different meals can be made from the set menu?

MATRIC DANCE MENU
STARTERS
Greek Salad
Fried Squid Heads
MAINS
Fillet Steak served with Mushroom Sauce and Vegetables
Fish of the Day with Lemon Butter Sauce
Vegetable Lasagne
DESSERT
Ice-cream and Hot Chocolate Sauce
Slice of Baked Cheese Cake

Factorial notation:

The factorial (!) of a natural number n is the product of the positive integers less than or equal to n .

$$4! = 4 \times 3 \times 2 \times 1$$

$$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$n! = n \times (n - 1) \times (n - 2) \times (n - 3) \times (n - 4) \times \dots \times 1$$

$$0! = 1$$

Example: In how many different ways can the letters of the word EIGHT be arranged (repetition of letters are not permitted)?

Example: @ different Science books, four different Afrikaans books and two different Maths books are placed on a shelf.

- a) How many different ways can they be arranged?

- b) How many ways can they be arranged if the books of the same subject must be placed together?

Dealing with repetition:

Consider the word DAD. In the previous examples if there were 3 letters in a word there were $3!$ ways of arranging the letters. However in all of these cases there were no repeated letters. In the case of the word DAD if the D's were swapped the word would still be DAD and it would not count as a different arrangement. In order to account for the repetition of letters, we divide $3!$ by $2!$ because the letter D is repeated twice. If the word was DADDY then we would divide $5!$ by $3!$ As the letter is repeated three times.

Example: Given the word PROBABILITY:

- a) Determine the number of different ways that the letters can be arranged.

- b) What is the probability the word will start with a B?

Selections:

We will now use factorial notation to deal with problems where there are a limited number of places which have to be filled from a group.

If r spaces have to be filled from n number of different choices, the number of possible arrangements are:

$$\frac{n!}{(n-r)!}$$

Example: There are 8 contestants in a race. How many different ways can 1st, 2nd and 3rd place be arranged?